

Suspension Balance & Lap Time Simulation in AVL VSM

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Task 1: Steady state cornering

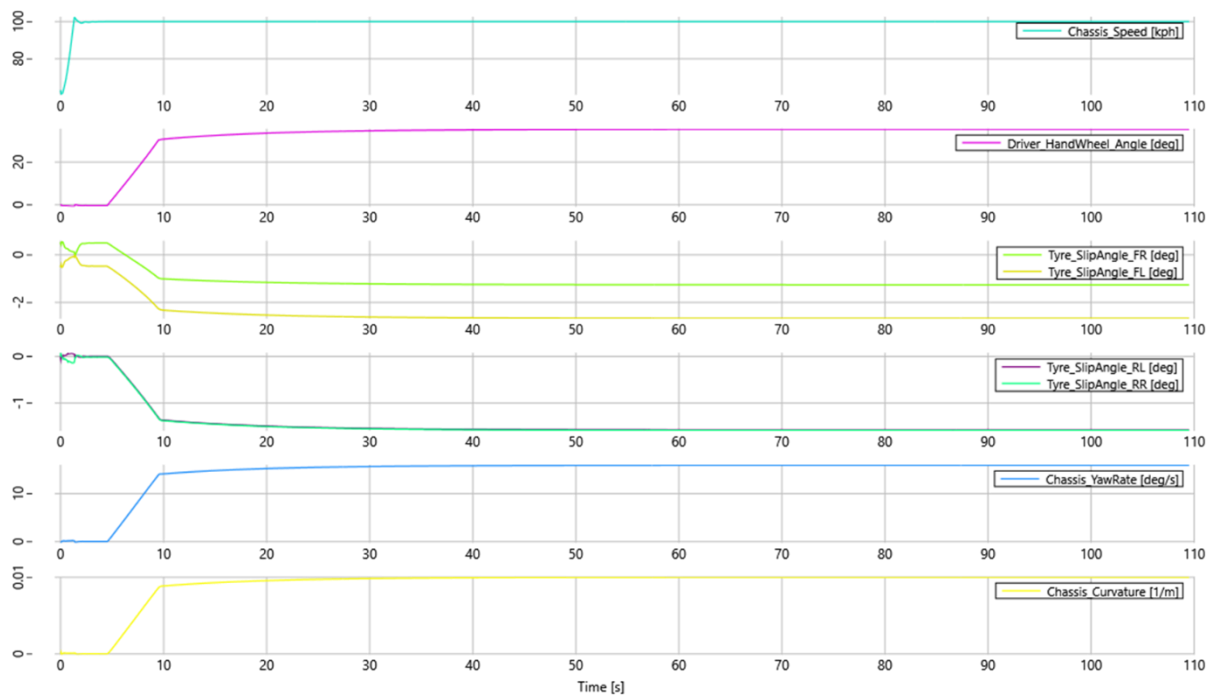
Perform simulations in VSM using the standard sports sedan and generate understeer, neutral balance and oversteer with the following conditions:

- Turning radius 100m
- Speeds: 100 and 150 km/h

Changeable car parameters:

- COG position
- Springs
- Dampers
- ARB-s
- Ride height
- Camber angle
- Toe angle
- Aero package

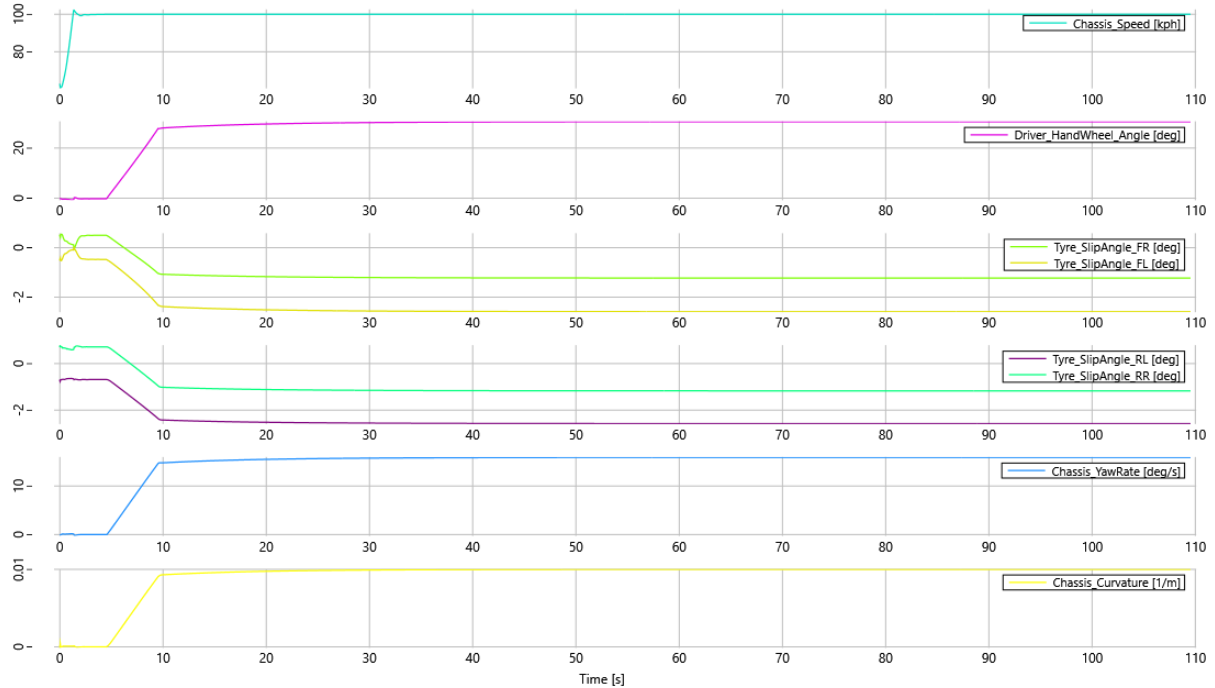
With the default setup the sport sedan generates **understeer**, because the front slip angle's absolute average (1,96) is higher than the rear's absolute (1,575) → 0,385.



1. Figure: Understeer setup @ 100 km/h

Default car setup → Understeer @ 100 km/h								
	COG pos	Springs	Dampers	ARB-s	Ride height	Camber angle	Toe angle	Aero package
Front	45	45 N/mm	default	19 N/mm	130 mm	-1	-0,5	1
Rear	55	55 N/mm	default	14 N/mm	130 mm	-1,5	0	1

For the purpose of having a neutral steer car, I have increased the stiffness of the rear axis on the default car setup. With the change of the toe angle and the ARB stiffness we can have a more balanced setup.

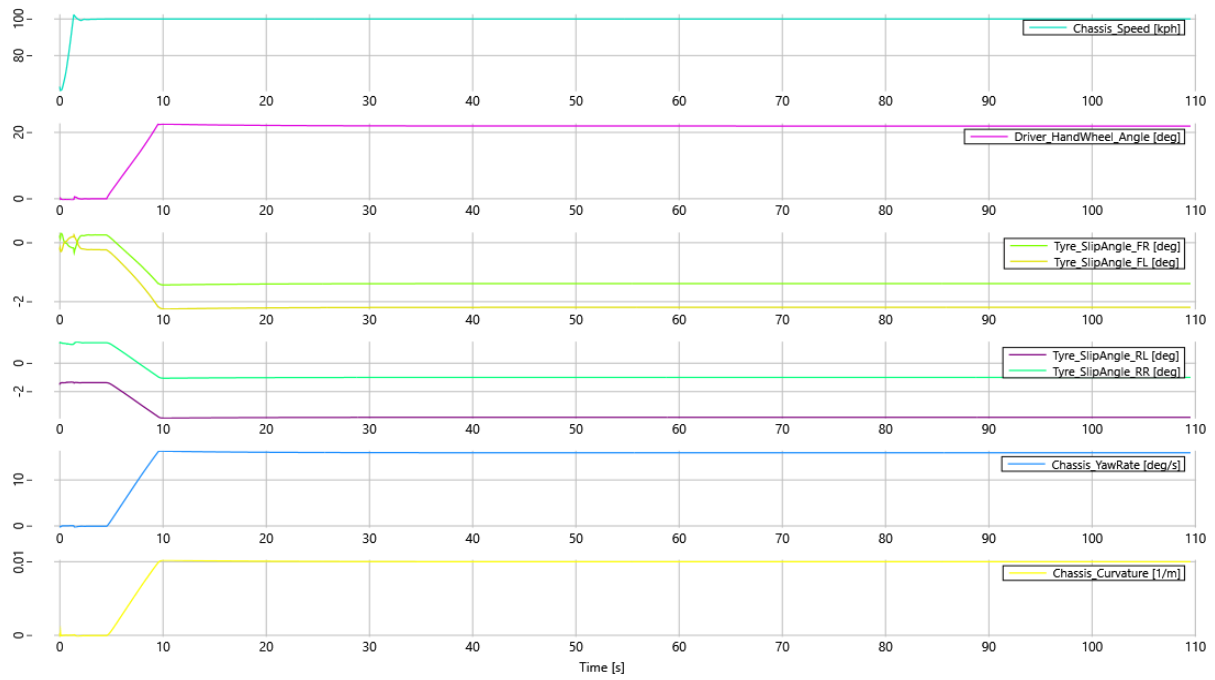


2. Figure: Neutral steer setup @ 100 km/h

Neutral steer @ 100 km/h								
	COG pos	Springs	Dampers	ARB-s	Ride height	Camber angle	Toe angle	Aero package
Front	45	45 N/mm	default	14 N/mm	130 mm	-1	-0,5	1
Rear	55	55 N/mm	default	30 N/mm	130 mm	-1,5	-0.7	1

With these changes the absolute delta slip angle between the front and rear is 0,035 which can be considered a **neutral-steer** car.

In order to have an oversteer car setup, we must have higher slip angles at the back than at the front. To achieve this, I have stiffened the rear end more, adjusted toe and raised the ride height at the rear, and lowered its oversteer at the front.



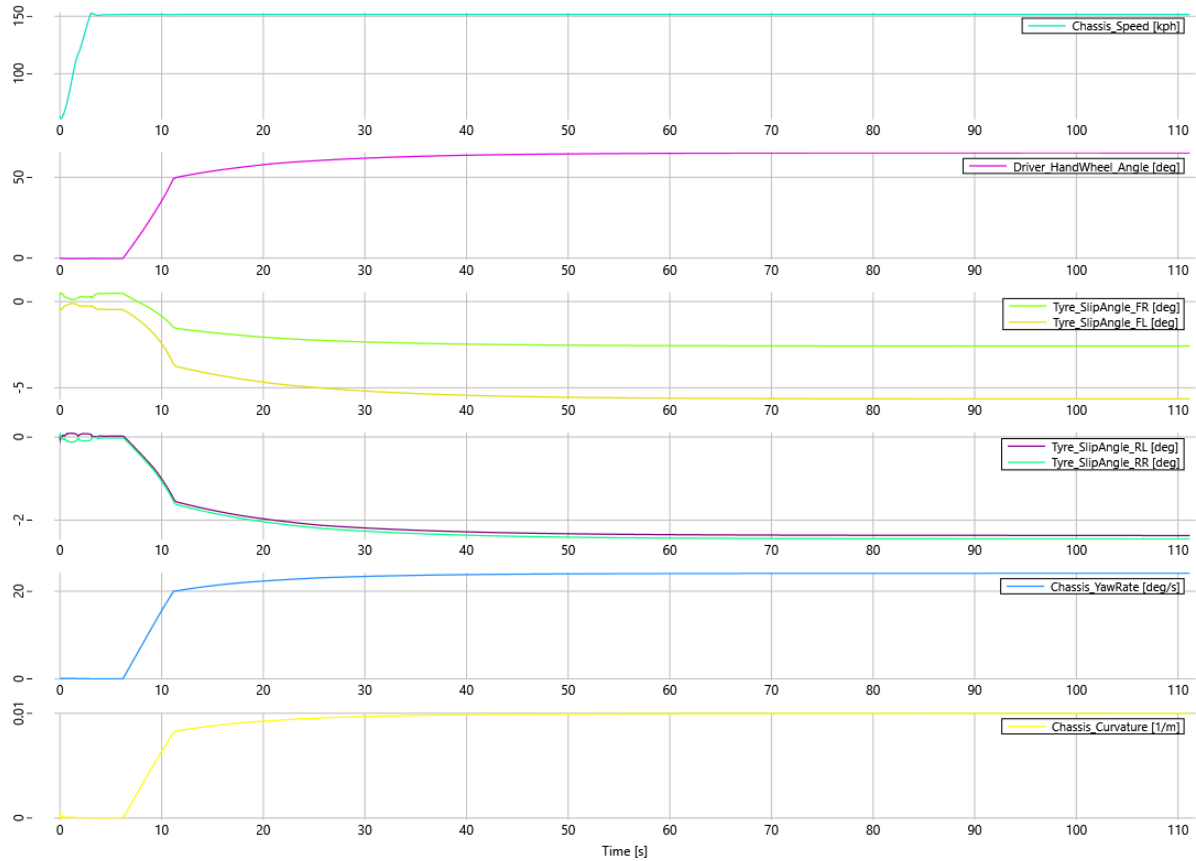
3. Figure: Oversteer setup @ 100 km/h

Oversteer @ 100 km/h								
	COG pos	Springs	Dampers	ARB-s	Ride height	Camber angle	Toe angle	Aero package
Front	45	45 N/mm	default	10 N/mm	120 mm	-1	-0,2	1
Rear	55	55 N/mm	default	40 N/mm	185 mm	-1,5	-1,4	1

With these changes the absolute delta slip angle between the front and rear is 0,615 which can be considered an **oversteer** car.

Steady state cornering at 150 km/h

To be able to use the VSM simulation software at this speed, I had to increase the full aero package at the front and at the rear as well, and increased Lat Grip Factor for the four tyres from 100% to 200%, otherwise the simulation couldn't finish.

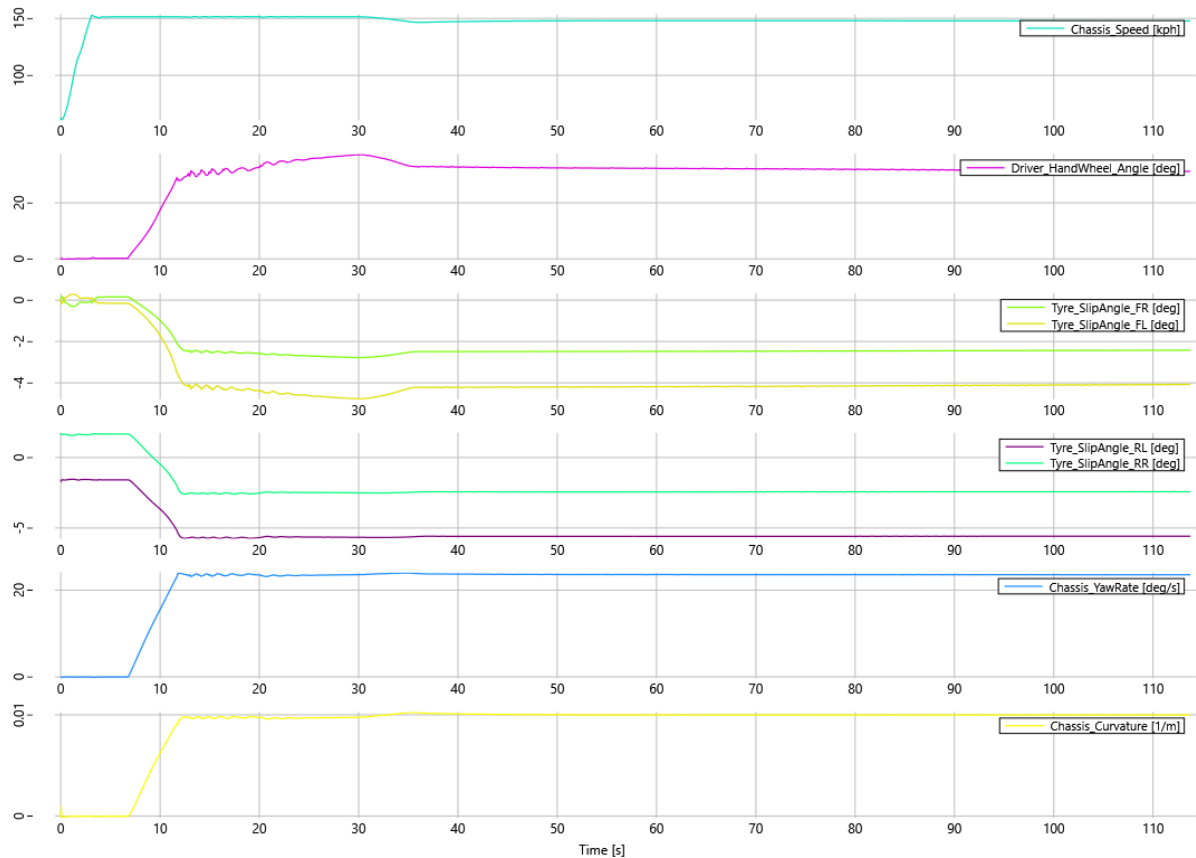


4. Figure: Understeer setup @ 150 km/h

Understeer @ 150 km/h								
	COG pos	Springs	Dampers	ARB-s	Ride height	Camber angle	Toe angle	Aero package
Front	45	45 N/mm	default	16 N/mm	130 mm	-1	-0,5	2
Rear	55	55 N/mm	default	22 N/mm	130 mm	-1,5	0	2

With the ARB and aero package adjusted for the 150 km/h case, the car is heavily understeered — the absolute delta slip angle between the front and rear is 1.7.

At first, I wanted to achieve a neutral state car from the previous setup, but I have changed too many parameters, and too aggressively which caused the car to be heavily oversteered, but after some fine tuning I achieved the following results:

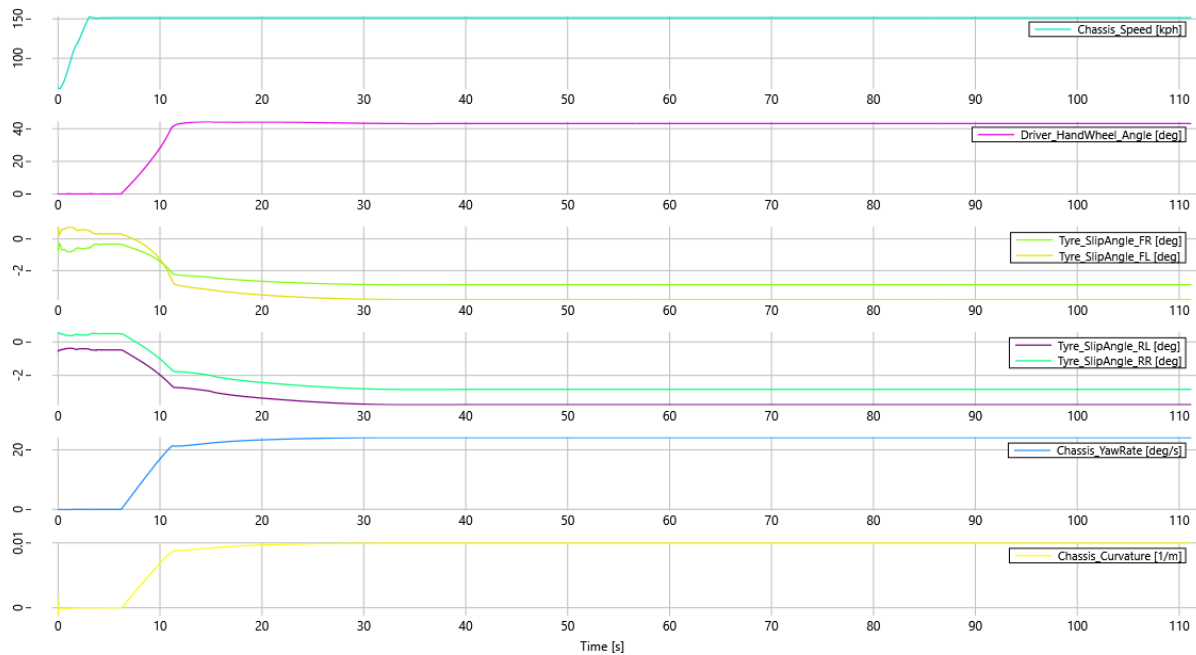


5. Figure: Oversteer setup @ 150 km/h

Oversteer @ 150 km/h								
	COG pos	Springs	Dampers	ARB-s	Ride height	Camber angle	Toe angle	Aero package
Front	45	46 N/mm	default	16 N/mm	120 mm	-1	0	2
Rear	55	55 N/mm	default	22 N/mm	180 mm	-1,5	-1,6	2

With these changes the absolute delta slip angle between the front and rear is 0,73 which can be considered an **oversteer** car. The diagram shows that, in this case, the virtual driver cannot achieve a smooth corner exit because the steering angle is not continuous.

In order to have a neutral state car, a lot of parameters had to be fine-tuned. After a period of testing, it became clear that toe angle has the largest effect on the front-rear balance, but only up to a certain point – beyond that, further changes stopped making a meaningful difference.



6. Figure: Neutral steer setup @ 150 km/h

Neutral steer @ 150 km/h								
	COG pos	Springs	Dampers	ARB-s	Ride height	Camber angle	Toe angle	Aero package
Front	45	60 N/mm	default	16 N/mm	140 mm	-1	0,7	2
Rear	55	25 N/mm	default	22 N/mm	140 mm	0	-0,5	2

With these changes the absolute delta slip angle between the front and rear is 0,065 which can be considered a **neutral steer** car.

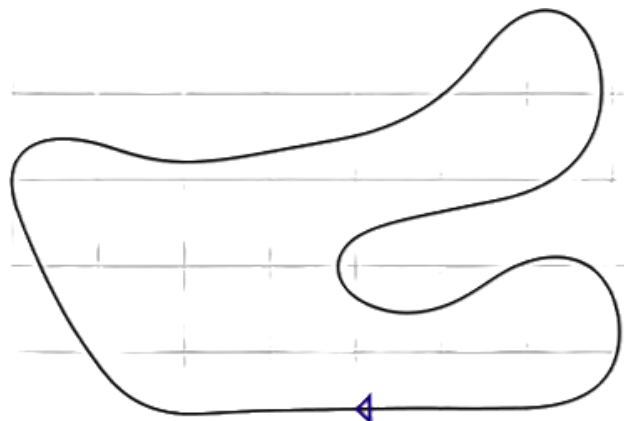
Task 2: Lap driving

Perform simulations in VSM using the standard sports sedan with the following conditions:

- Track: Hockenheim Short
- Achieve a fast lap

Changeable car parameters:

- COG position
- Springs
- Dampers
- ARB-s
- Ride height
- Camber angle
- Toe angle
- Aero package

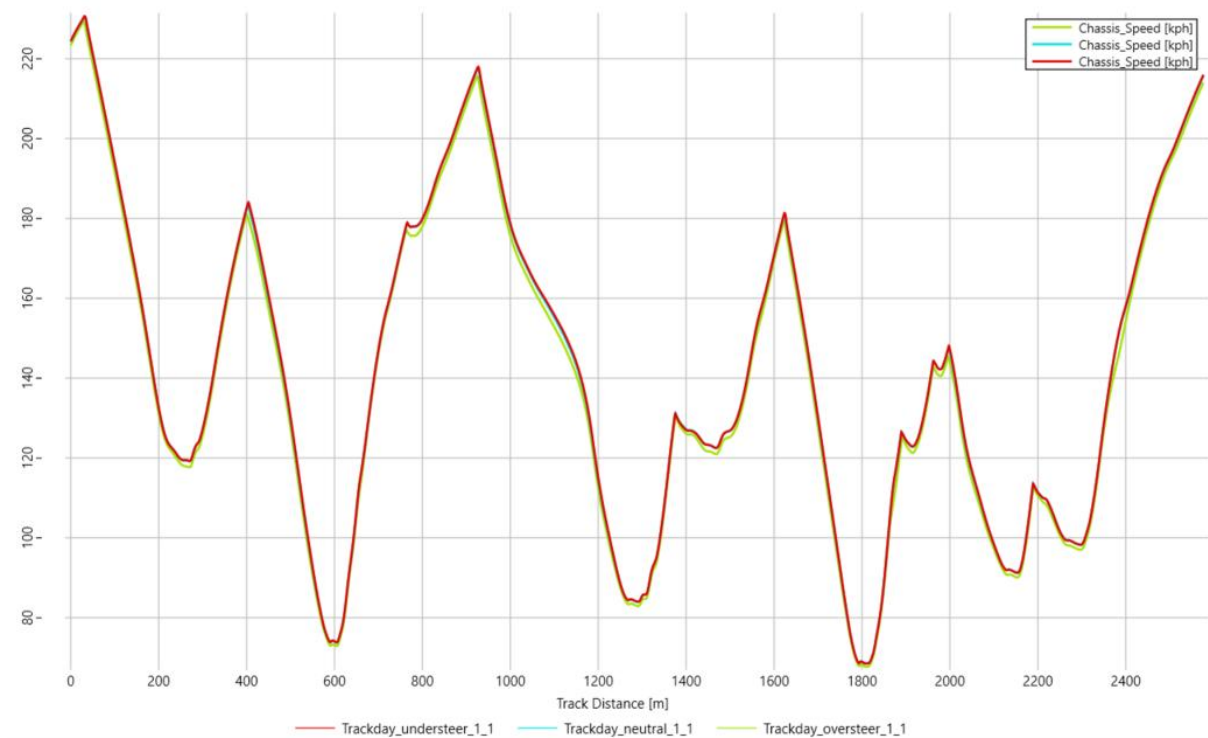


7. Figure: Hockenheim Short track layout

For this task, I used the models developed for the 100 km/h steady-state cornering case. This provided a setup for each vehicle type, allowing me to analyse which configuration performs best on the Hockenheim short track within the VSM simulation and given parameter range.

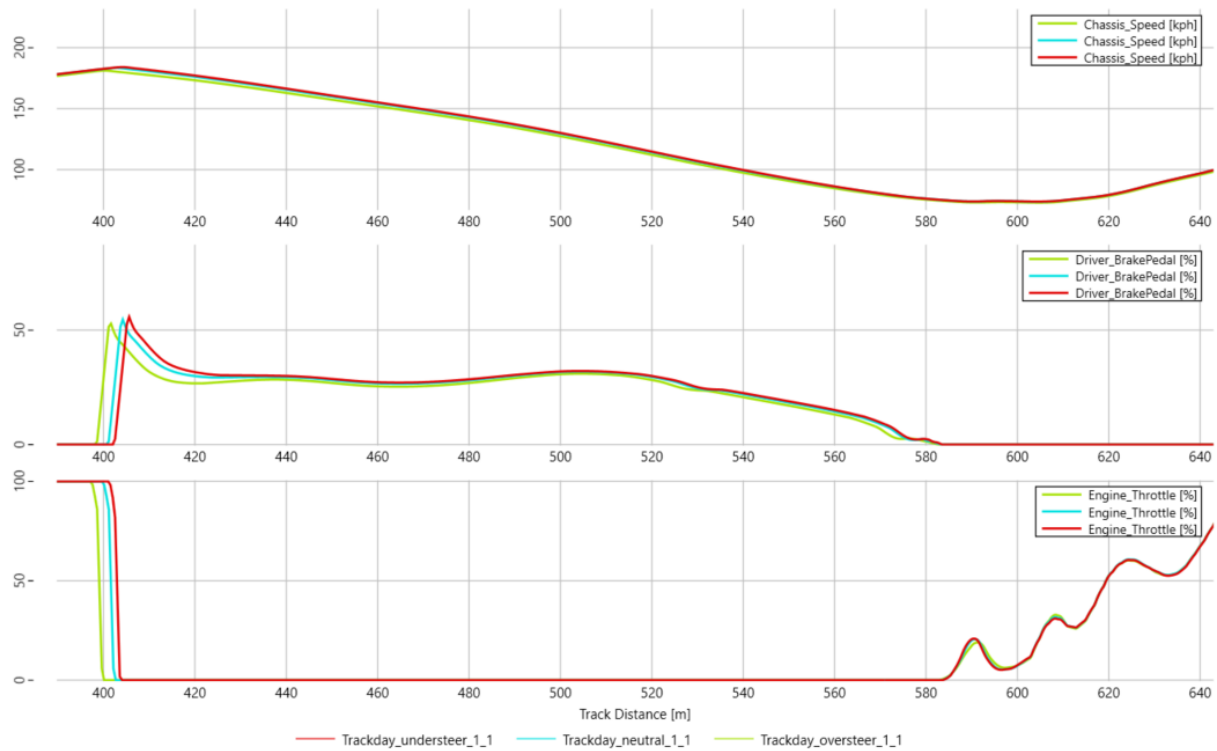
Default car setup → Understeer @ 100 km/h								
	COG pos	Springs	Dampers	ARB-s	Ride height	Camber angle	Toe angle	Aero package
Front	45	45 N/mm	default	19 N/mm	130 mm	-1	-0,5	1
Rear	55	55 N/mm	default	14 N/mm	130 mm	-1,5	0	1
Neutral steer @ 100 km/h								
	COG pos	Springs	Dampers	ARB-s	Ride height	Camber angle	Toe angle	Aero package
Front	45	45 N/mm	default	14 N/mm	130 mm	-1	-0,5	1
Rear	55	55 N/mm	default	30 N/mm	130 mm	-1,5	-0.7	1
Oversteer @ 100 km/h								
	COG pos	Springs	Dampers	ARB-s	Ride height	Camber angle	Toe angle	Aero package
Front	45	45 N/mm	default	10 N/mm	120 mm	-1	-0,2	1
Rear	55	55 N/mm	default	40 N/mm	185 mm	-1,5	-1,4	1

With these setups the **fastest car was with understeer**, which completed the circuit in **71.53** seconds, then the **neutral steer** car followed with **just a 0,12 (71.65s) second difference** and the **oversteer setup had more than a second gap, finishing in 72.54** seconds.



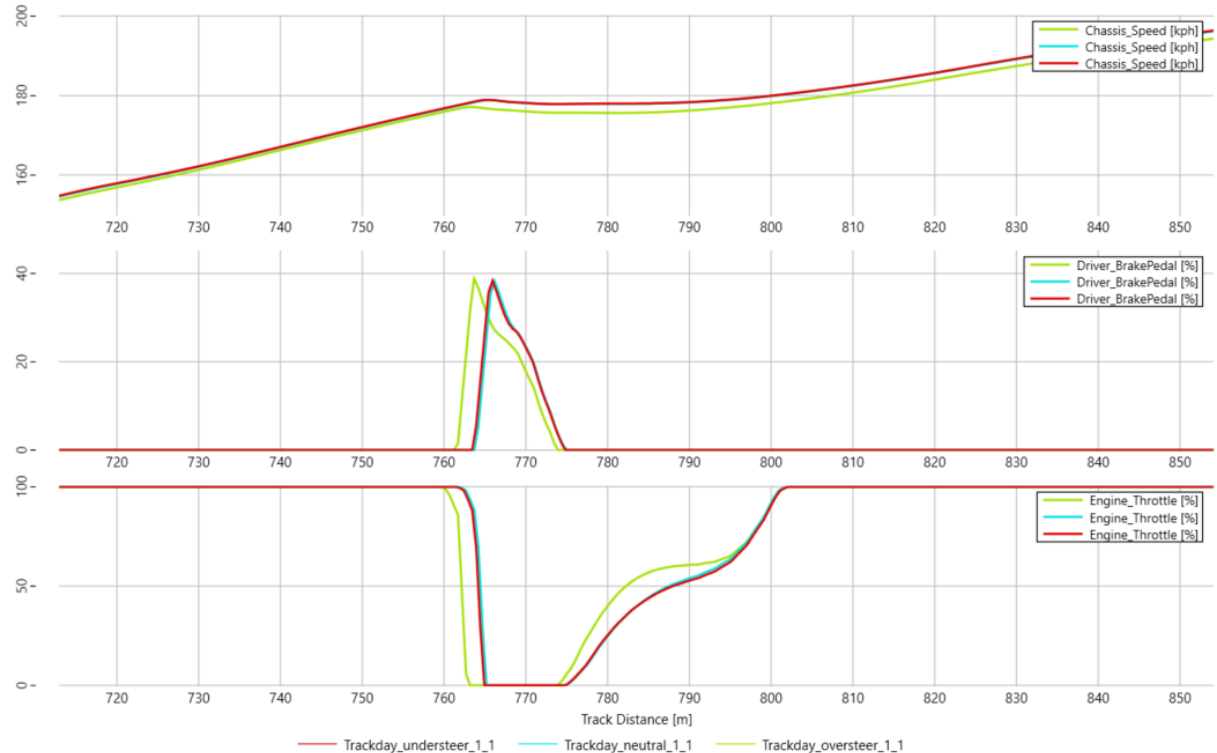
8. Figure: Lap time results

One of the most influential parts of the track is shown on the following picture:



9. Figure: T2 comparison for different setups

In this section, it can be seen that the oversteer-biased setup has to brake earlier, which reduces the car's pace. Additionally, it prevents a well-controlled corner exit, resulting in further time loss. And as the end time results show, the understeered and neutral steered setup is close to each other during the whole track.



10. Figure: T2 exit and T3 comparison for different setups

Another crucial part where the oversteered car loses time is at the figure shown above. The oversteered car must brake earlier again. This lets it accelerate faster out of the corner, but it still can't reach as high a top speed as the other two setups.

Closing Summary for Task 2:

Across the Hockenheim short lap, the understeer and neutral-steer setups performed almost identically, while the oversteer setup lost more than a second overall. The earlier braking the oversteer-biased car needed into corners outweighed the extra acceleration it gained on exit, making it the slowest of the three across the full lap.